Chapter 48
On the Evaluation of BDD Requirements with Text-based Metrics: The ETCS-L3 Case Study

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Abstract A proper requirement definition phase is of a paramount importance in software engineering. It is the first and prime mean to realize efficient and reliable systems. System requirements are usually formulated and expressed in natural language, given its universality and ease of communication and writing. Unfortunately, natural language can be a source of ambiguity, complexity and omissions, which may cause system failures. Among the different approaches proposed by the software engineering community, Behavioural-Driven Development (BDD) is affirming as a valid, practical method to structure effective and non-ambiguous requirement specifications. The paper tackles with the problem of measuring requirements in BDD by assessing some traditional Natural Language Processing-related metrics with respect to a sample excerpt of requirement specification rewritten according to the BDD criteria. This preliminary assessment is made on the ERTMS-ETCS Level 3 case study whose specification, up to this date, is not managed by a standardisation body. The paper demonstrates the necessity of novel metrics able to cope with the BDD specification paradigms.

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48.1 Introduction

The Internet of Things (IOT) is changing not only the way devices interact, but also their development and design. Manufacturing is becoming more complex, consequently, elaborate hardware and software components (and their increasing connectivity) are becoming necessary. With the Requirement Engineering (RE) discipline, the complexity and interdependencies between the different components of a system can be overcome. RE is a complex body of knowledge that counts different subdisciplines from requirement elicitation to validation. Problems encountered in RE are often reported as the main cause of project failures both under technical and managerial aspects. In many worst cases, requirement ambiguities and non-completeness may bring to catastrophic failures with life loss and environmental disasters, on the other hand, forcing system analysts to use a formal, mathematically sound method, may increase time-to-market [1].

System Requirements Specification (SysRSs) are, usually, written in Natural Language (NL), which is intuitive and universal and, to be effective, they must be described in a specific, verifiable, clear and concise, accurate and comprehensible manner. While writing SysRSs in NL makes it easy to perform and interpret, on the other hand it eases to incur ambiguity. A good RE methodology is critical to the success of a project.

In recent years, behaviour-driven development (BDD) has gained the attention of researchers and developers, in both academia and software industry. BDD is an agile development approach originally developed by Dan North [2]. The main objective of the BDD is the transformation of requirements into working and verified code, as long as the requirement is easily understandable and clearly written. BDD provides a specific language that everyone can understand to assist stakeholders in specifying their tests [3, 4]. Recently, several Natural Language Processing (NLP) approaches have been developed to support requirements analysis. Automatic identification of terms, attributes, constructions in the SysRSs constitutes the main objective of these approaches [5, 6]. At the best of our knowledge, there are few attempts in the scientific literature to assess the quality of a SysRSs written according to BDD. Such attempts mainly focus on the definition of guidelines and checklists extracted from experts panels [7].

The main objective of this paper is to demonstrate that traditional metric-oriented ways of measuring requirements do not capture the intrinsic nature of BDD. Those methods are designed for a classical sentence-centred writing style and are hence based on NLP methods, this latter used, among other things, in supporting malicious activities and objects detection [8–11]. To have a first idea of how traditional methods fail, a railway requirement specification document has been analysed with HOMER.[1] The methods defined in this paper are applied on the case study of the European Train Control System-Level 3 (ETCS-L3) requirement specification, the next-generation signalling systems that promises to revolutionize the railway transportation industry by strongly increasing network capacity.

[1] https://github.com/wyounas/homer.

The remaining sections are organized as follows. Section 48.2 describes the main approaches existing in the scientific literature to assess quality of SysRSs. Section 48.3 presents the proposed methodology to determine the features of a SysRSs document, while Sect. 48.4 describes its application to the railways case. Analysis results are discussed in Sect. 48.5. Conclusions are drawn in Sect. 48.6.

48.2 Related Works

The main contribution given by the research community on how to write a valuable SysRSs document concerns the identification and classification of ambiguity-related defects in requirements [12–17]. The importance given to this topic arises from the consciousness of the strong relation between an almost ambiguity-free document and the early detection of errors in developmental cycle of a system. It is a smart technique to prioritize the definition of the expected behaviours and requirements of the system and to use them to revise the project step-by-step.

Some of the previously mentioned papers also define tools for the detection of ambiguity in requirement documents. Fabbrini et al. define a tool called *QuARS (Quality Analyzer for Requirement Specifications)* able to perform an analysis of NL requirements in a systematic and automatic way by means of NLP techniques [15, 18]. Génova et al. propose a tool called *RQA (Requirements Quality Analyzer)*, based on the *QuARS* tool [19]. Instead, Arora et al. deal with the necessity of using requirement templates and provided the tool *RETA (REquirements Template Analyzer)* to check the conformance to specific templates [20].

Most of the works present in the literature focus on the assessment of tools [18, 20], thus, using the class of defects they considered most important, they evaluated the efficiency of the tools in correctly labelling as accepted or rejected the requirements. The used approach is mainly based on the evaluation of recall, precision, accuracy and, in some cases, of the F-measure [21, 22]. However, this approach involved, usually, a domain expert which manually labelled the requirements and compared them to the results obtained by the tool.

Among the others, it is worth highlighting the work of Davis et al., in which the authors carefully examine the attributes considered relevant when writing a SysRSs [13]. Moreover, for each of the attributes, they suggested a metric to use in measuring and improving the quality of the requirements and, for almost all of them, they provided a recommended weight relative to the other attributes. Finally, Ferrari et al. proposes an interesting case study, where the railway domain is considered, and the *QuARS* tool is used [16].

48.3 Methodology

The proposed methodology involves the dataset definition and preparation, text pre-processing, features definition and extraction, and model extraction.
Dataset: three datasets are considered. A *golden* dataset of traditionally expressed requirements, a second *target* dataset and an *improved* one, obtained by manually translating the requirements of the target dataset according to the Gherkin notation and BDD approach, are evaluated. All the datasets consist of a list of pairs `(id, sentence)`.
Text Preprocessing: a preprocessing phase is required to catch the different lexical features of the text under examination. The NLP text preprocessing is composed by: tokenization, lemmatization and Part of Speech (POS) tagging [23]. These phases can be computed through specific libraries which are part of the Natural Language Processing Tool Kit (NLTK) suite,[2] an open source library in Python Language. Each word in a sentence is the atomic unit necessary for the fulfilment of the meaning of the sentence itself. Just as grammar analysis breaks the links between these words, to understand them and process them separately, so the tokenization phase applies the word_tokenize() function to transform each word into an isolated *token* to categorize it. The next step emulates the Morphological Analysis (MA) that deals with the structural study of words. MA aims at identifying words with the same lexical form, recognizing the different forms (plural, past tense of verb) and mapping them to the basic lexical form (singular, infinitive form of verb). This process is the opposite of tokenization because it associates rather than dissociates, but it is fundamental as it allows the recognition of ontological concepts [24]. NLTK provides WordNetLemmatizer() which uses the WordNet database to search for word headings. Through lemmatization, each token is mapped to its basic form, i.e., converted into *lemma* in the WordNet dictionary. The last process is the classification of each token into its own lexical and functional category via NLTK's pos_tag() function. Each token is assigned a POS tag and, if necessary, can be assigned a fine-grained tag determined by the morphology.[3] So through the post tagging phase each word of the initial sentence will be labeled, for example, as a noun (NN), verb (VB), conjunction (IN), cardinal number (CD), punctuation mark (PUNCT) or more.

Features Extraction: according to Fig. 48.1, given *tagged words* for all the requirements, it is now necessary to define the characteristics to be highlighted. Starting from the *ambiguity detection rules* defined in [25], the work presented here has extended this list by considering a merge with the *patterns* expressed in [16] obtaining 19 features summarized in Table 48.1.

Model Extraction: to extract a single overall score from the defined feature set, our methodology considers tuning some linear weights on the *golden* dataset. During this phase, the scalar product between the features vector and a random weight vector is minimized. The minimization is accomplished by means of standard bounded and

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[2] https://www.nltk.org.

[3] https://universaldependencies.org/u/pos/.

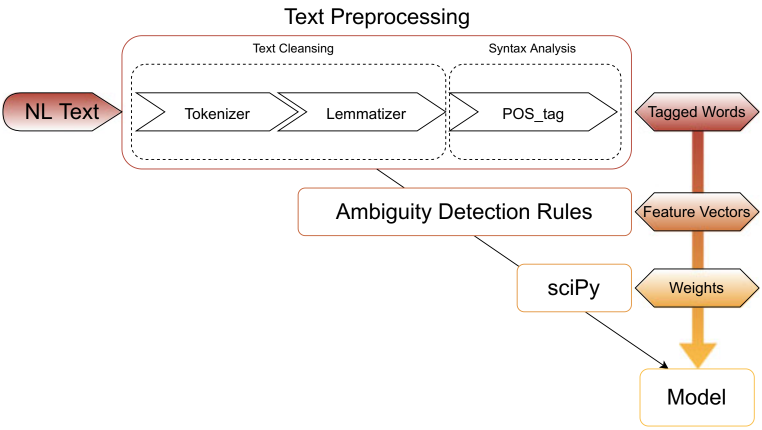

Fig. 48.1 Methodology Pipeline

constrained optimization algorithms, supported by the sciPy library.[4] Such weights are used during the evaluation (on both the target and improved datasets) to compute the overall score as the scalar product between the feature values and the weights, as they are computed during tuning.

Feature Extension: to improve the considered feature set, further NLP-based techniques are applied to the feature set to try a first adaptation to the BDD methodology. Some of the practical adopted solutions are: excluding "*Given*" from the nouns and verbs counts and excluding "*Given*", "*When*", "*Then*" from the count of total words and the Upper Case ones.

48.4 The ERTMS-ETCS Level 3 Case Study

From a general perspective, a Control, Command and Signalling (CCS) system is in charge of achieving safe trains movement. To pursue safety, railroads have been commonly divided into fixed track sections, known as fixed blocks. Fixed blocks reflect the choice of developing CCS on hardware intrinsic fail-safe technologies, supported by the existence of physical circuits able to determine the presence of the train inside a track area. The European Railway Traffic Management System (ERTMS) is one of the first "global" attempts to a CCS softwarization, with the main goal of ensuring interoperable and safe signalling across Europe. Computer-based systems and communication technologies allow a software-based control of the train. The European Train Control System (ETCS) is the signalling element of ERTMS that includes the control of movement authorities, the automatic train protection and the interface to interlocking. Today, ETCS is present in Europe and worldwide in

[4] https://scipy.github.io/devdocs/tutorial/optimize.html.

Table 48.1 Evaluation features

Name	Description	Type
NN before NNP	There is at least one noun before a pronoun	1 (T), 0 (F)
Conjunction	Number of conjunctions	Integer
>1 PRP	There is more than one pronoun	1 (T), 0 (F)
isThereNN	There are at least two singular nouns	1 (T), 0 (F)
>4 VB	Number of verb is greater than four	1 (T), 0 (F)
>2 punct.	Number of punctuation marks is greater than two	1 (T), 0 (F)
Noun	Nouns count	Integer
Pronoun	Pronouns count	Integer
Verb	Verb count	Integer
Punctuation	Punctuation marks count	Integer
isThereCCorPRP	There is one relative clause and one pronoun together	1 (T), 0 (F)
>1 CapLet.	There are at least two capital letters	1 (T), 0 (F)
>2 IN	There are at least two prepositions	1 (T), 0 (F)
isThereNNSorNNPS	There are at least two plural nouns	1 (T), 0 (F)
Anaphoric	Anaphoric words count	Integer
Word counter	Number of words for sentence	Integer
Vague word	Vague words count	Integer
Adverbs	Modal adverbs count	Integer
Passive verb	Passive verb count	Integer

two possible configurations: ETCS Level 1 and ETCS Level 2, still based on fixed blocks. These two levels are well specified in a complex set of specifications; among them, the SUBSET-026 version 3.0.0 [26].

With the ETCS-L3 initiative, railway stakeholders envisage implementing a promising train control system: the Moving Block (MB). Eventually, train position detection will be not based on physical track circuits, enabling a considerable increase in network capacity. Up to this date, ETCS-L3 is still in the research phase and is the subject of different EU-funded research project involving both industry and academia. To name a few: X2Rail-3,[5] ASTRail[6] and PERFORMINGRAIL.[7] Only few pilot lines have been built and used for experimenting technologies and solutions; international organizations have not still started the standardization process.

To this aim, and according to the methodology defined in Sect. 48.3, the comparison is made on the basis of the following documents:

1. SUBSET-026 v3.0.0 part 5: reporting the main system requirements of the procedures of ETCS Level 2 [26]. The first specification is an assessed document; it has been refined during different formal revisions and feedback from concrete system construction and operation. A requirement example is the 5.12.2.5: *When the driver closes a desk and opens the other one of the same engine, the ERTMS/ETCS on-board equipment shall be able to calculate the new train position data (train front position, train orientation), by use of the previous data.*
2. X2Rail-3 D4.2 part 3: containing the tentative specification for the ETCS-L3 [27]. The document constitutes a part of a public deliverable in the EU research project X2Rail-3, run by the main European industrial companies involved in railway signalling and systems. A requirement example is the REQ-Reserved-2: *Following a request from the TMS, the L3 Trackside shall be able to reduce or remove an existing Reserved Status Area of track only after adjusting any given authorization accordingly.*

In this context, a part of the SUBSET-026 v3.0.0, about 52 requirements, is considered as the *golden* dataset; X2Rail-3 D4.2 is the *target* dataset, containing 142 requirements. All these last 142 requirements are then manually re-written according to the Gherkin notation, constituting the *improved* dataset. An example of this activity is reported here, with the rewriting of REQ-Reserved-2: ***Given*** *the L3 trackside on* **When** *the TMS sends a request* **and** *the L3 Trackside adjusts any given authorisation accordingly* **Then** *the L3 Trackside shall be able to reduce or remove an existing Reserved Status Area of track only after adjusting any given authorisation accordingly.*

48.5 Results

As a preliminary activity, the list of vague words used by the HOMER tool has been extended with other words.[8] Then, HOMER has been applied to both the considered documents [26, 27], showing us some generic insights: reading time in minutes, readability scores (Flesh reading ease and Dale-Chall readability scores), total paragraphs, sentences and words, average sentences for paragraph, average words per sentence and number of vague word (see Table 48.2).

[5] https://projects.shift2rail.org/s2r_ip2_n.aspx?p=X2RAIL-3.

[6] http://www.astrail.eu/.

[7] https://www.performingrail.com/.

[8] https://www.blinn.edu/writing-centers/handouts-and-worksheets.html.

Table 48.2 Results of HOMER execution

	Subset-026	X2Rail-3
Flesch reading ease	Very difficult	Difficult
Dale-Chall readability score	7–8th grade	4th grade
Paragraphs	399	652
Avg sentences per paragraph	2.27	2.49
Sentences	904	1623
Avg words per sentence	20.7	16.68
Vague words	126	117

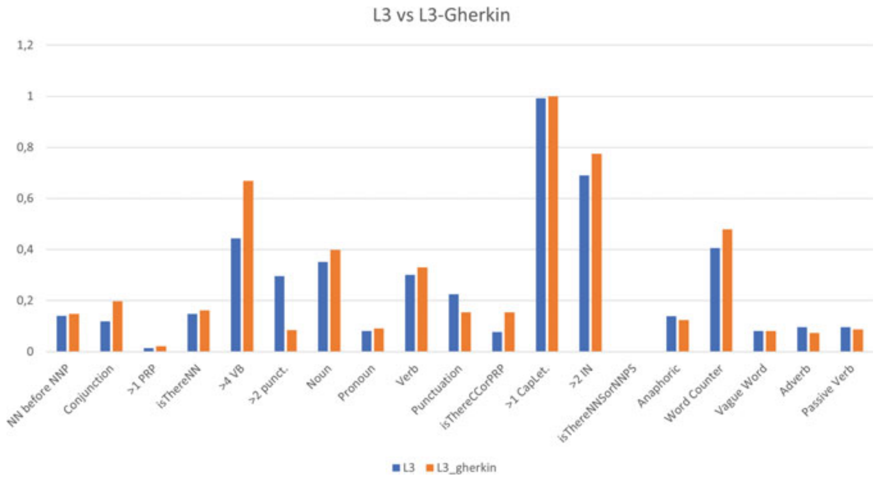

Fig. 48.2 Comparison between original and modified requirements

Then the tool, developed according to the methodology reported in Sect. 48.3, has been applied on the three datasets considered in Sect. 48.4. Figure 48.2 reports a graph illustrating the scaled measures for each considered feature for both the L3 requirements reported in [27] and a manual attempt to report such requirements in the Gherkin notation. Furthermore, the overall scores of such set of requirements are respectively 3.73908E−10 and 3.88056E−10.[9]

As it is possible to note, notwithstanding the rewriting of the requirements of L3 according to the Gherkin approach, the overall score of L3-Gherkin worsens. The single measures that most affect such increase are: >4VB (i.e., more than four verbs), >2IN (i.e., more than two conjunctions), word counter (i.e., the number of the words). Structuring better the requirements is not promoted but punished by quantitative evaluation.

[9] Lower score means better results.

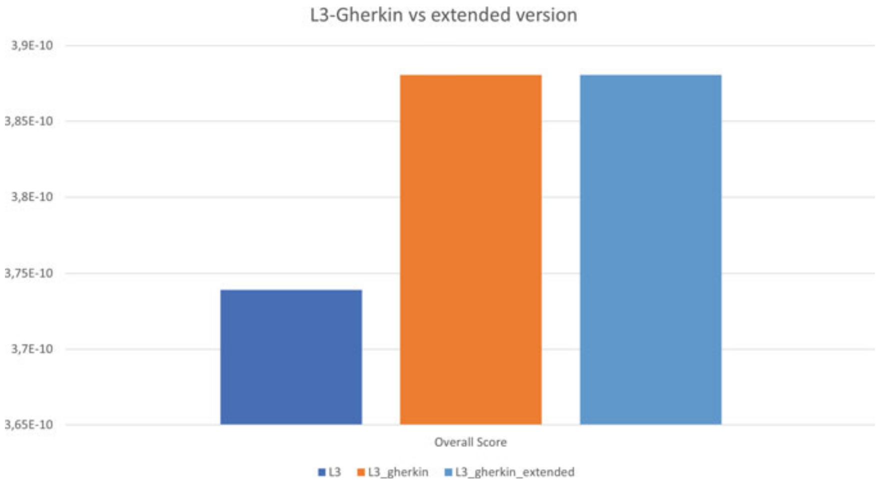

Fig. 48.3 Overall score on considered datasets

With respect to the *target* dataset, the *improved* dataset has the following characteristics. First, an increased number of conjunctions, words, verbs and nouns is observed. This increase is justifiable, as BDD templates lead to break complex sentences into simpler ones reducing punctuation and passive verbs; conversely, simple sentences increase the number of objects, active verbs, predicates and the conjunctions "and"/"or". The increase of the number of nouns and active verbs implies an improvement in the clarity of BDD sentences, as each verb has an explicit subject and action. The number of the total used words increases, accordingly. This is not negative as the ratio between the numbers of all the words and vague words do not imply a loss of clarity; indeed, a decrease in anaphoric words, adverbs, passive verbs and punctuation improves readability.

As a final consideration, there is the strong need of a novel approach to quantitatively measure requirements according to BDD. Traditional methods are not able to capture the advantages of such methodology that international expert panels agree to define one of the most promising techniques in RE [28]. In fact, also by applying the extended feature set (see Sect. 48.3) the mean overall score of the Gherkin-shaped requirements is still the same (see Fig. 48.3).

48.6 Conclusions

The present paper tackles with the problem of evaluating quantitative scoring of requirements in case of the adoption of the BDD methodology. This still constitutes an open problem since, notwithstanding a general improvement of the requirement comprehensibility, traditional NLP-based metrics and indices worsen: e.g., an increase of conjunctions, words, verbs and nouns was observed in the analysis of BDD-formatted requirements. This study constitutes a first research step oriented toward the definition of a more comprehensive set of criteria. The next research will be devoted to this direction, collecting different basic measures and generating a general score, also by means of Artificial Intelligence (AI) methods. As an example, a Named Entity Recognition (NER) process could be applied to the requirements to highlight the technical words of the specific domain under consideration. In this way, requirement specification activities could be integrated into Domain-Driven Design (DDD approaches in a closer way).

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